



Capstone Final Project

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OUTLINE



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- Methodology
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 - Dashboard
- Discussion
 - Findings & Implications
- Conclusion
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EXECUTIVE SUMMARY



- Analyze key insights and trends from the data
- Use data collection, wrangling, and explorations to create insightful visualizations
- Highlight trends in current and prospective years
- Discuss findings and implications of the trends of respondents
- Provide clear dashboard visuals to convey the data
- Discuss disparities in the data and analyze data quality
- Conclude the most important findings from the data set and key takeaways

INTRODUCTION



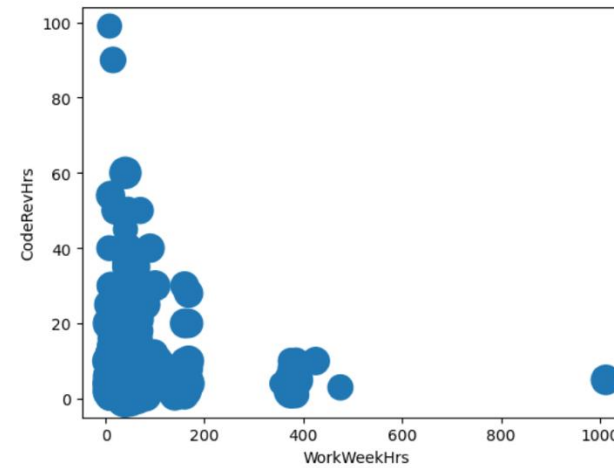
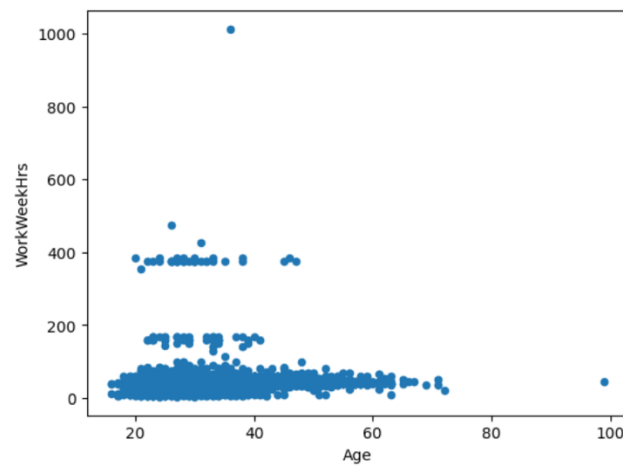
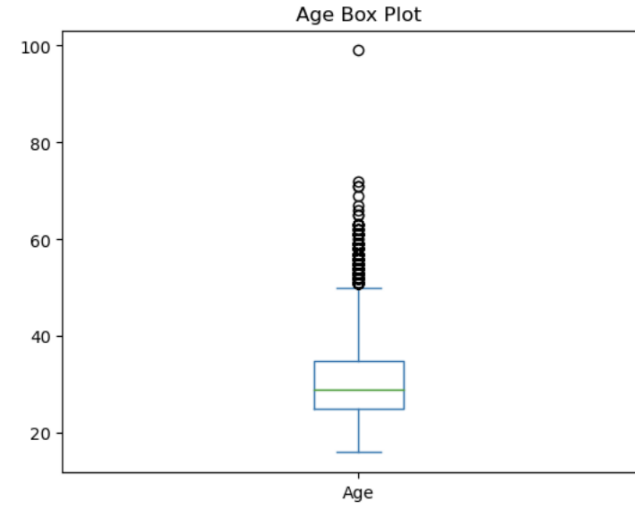
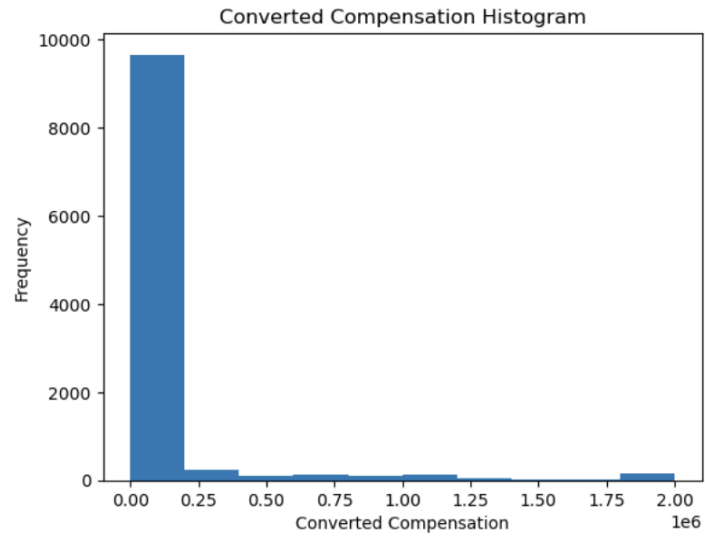
- We will aim to analyze the preferences of developers in current and futuristic periods
- Considering developer's backgrounds, we can make conclusions based upon various regions, industries, and job roles

METHODOLOGY



- Used Web APIs to scrape the internet for data collection on developer insights
- Cleaned the data using data wrangling techniques so that the data is normalized for analysis
- Explored the data for distributions, correlations, and finding outliers within USD normalized salaries
- Visualize the data relationships, and draw insights for comparisons
 - Highlight distribution, composition, and relationships of the respondents

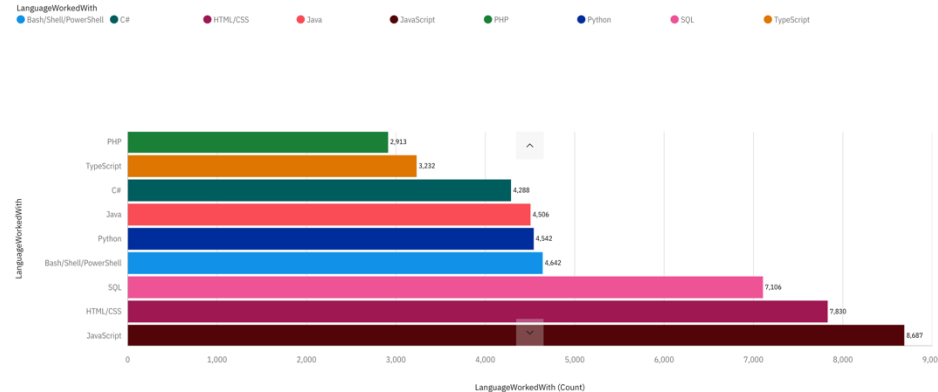
RESULTS



PROGRAMMING LANGUAGE TRENDS

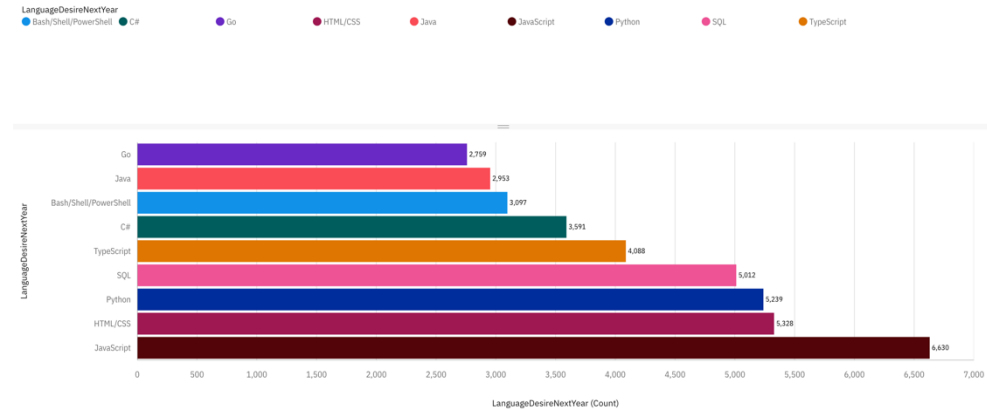
Current Year

Top 10 Languages Worked With



Next Year

Top 10 Languages Desired Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

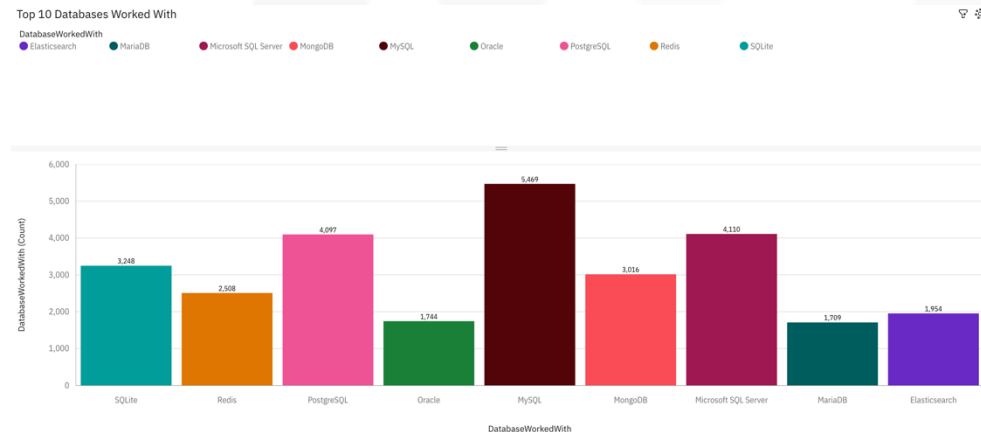
- Javascript remains the most desired language amongst participants
- HTML/CSS and SQL dropped by a large margin
- Go emerged on the list as a new top 10 language

Implications

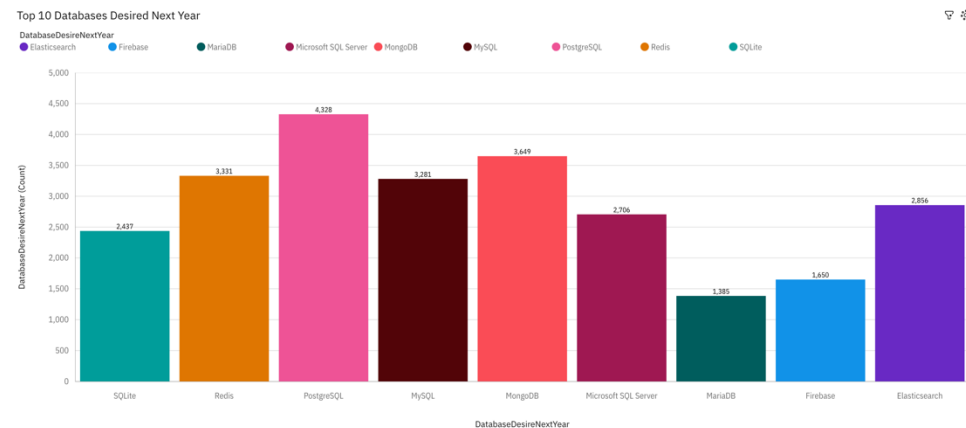
- Javascript is the leading development language
- Python gained popularity in the new market, reaching a similar amount of respondents as HTML/CSS and SQL
- New market trend allow for Go as a competitive choice

DATABASE TRENDS

Current Year



Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- MySQL dropped from 5,469 to 3,281 respondents
- The gap from most to least popular databases is high for the current year but gets tighter for future year projections
- PostgreSQL overtook MySQL as the most popular database

Implications

- New emerging technologies allow for the greater diversity among database usage
- The next years trend towards less variation
- PostgreSQL is the most desired future language

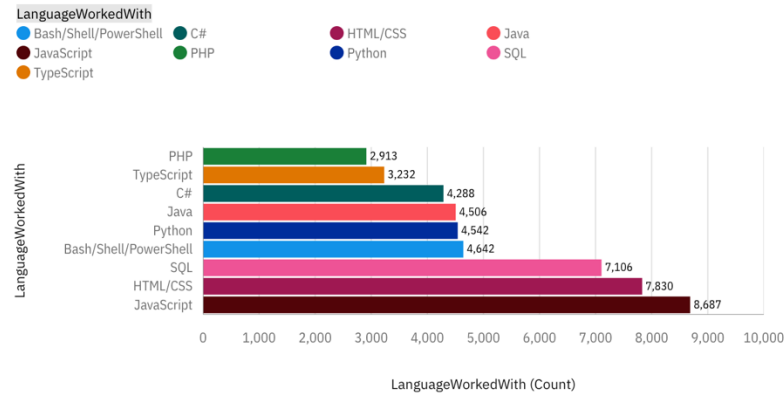
DASHBOARD



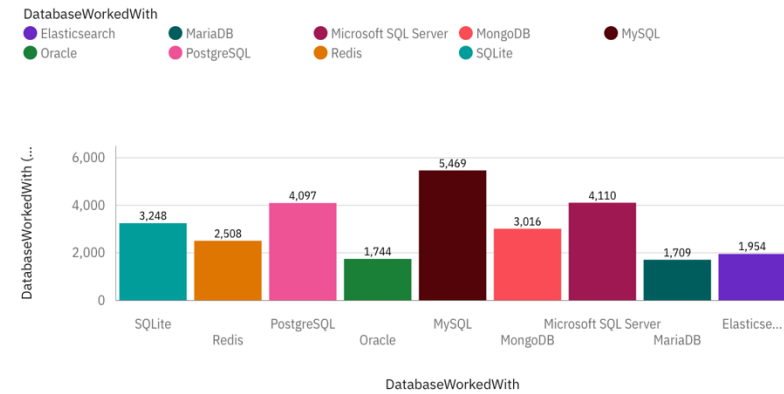
<https://github.com/haydencwiek/CapstoneDashboard>

DASHBOARD TAB 1

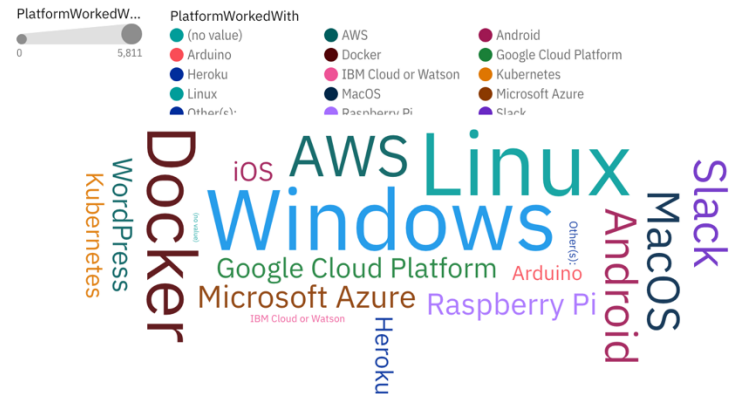
Top 10 Languages Worked With



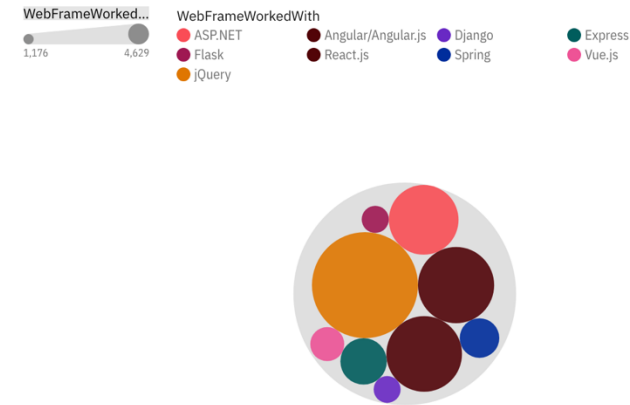
Top 10 Databases Worked With



Platforms Worked With

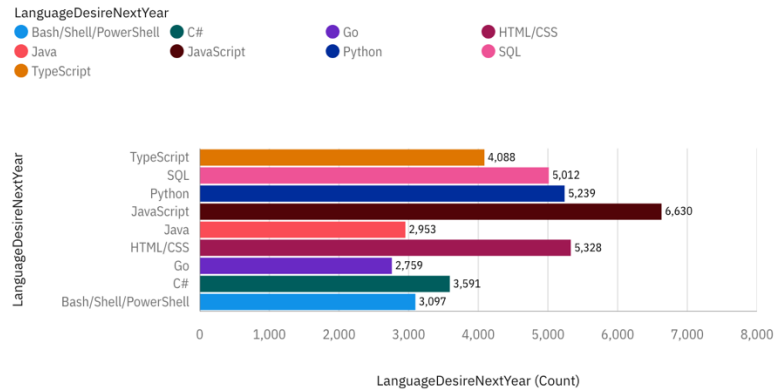


Top 10 Web Frames Worked With

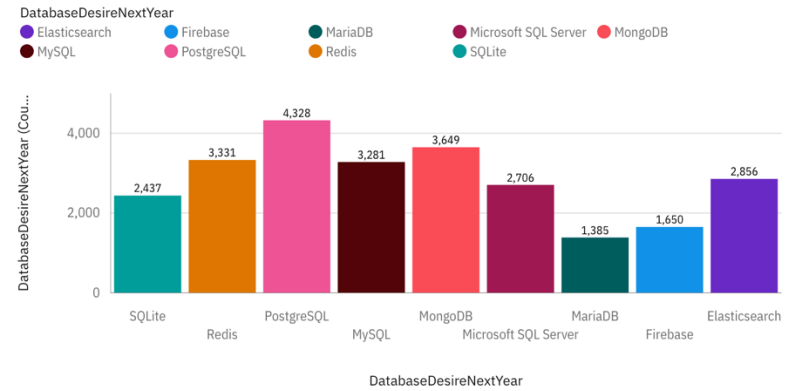


DASHBOARD TAB 2

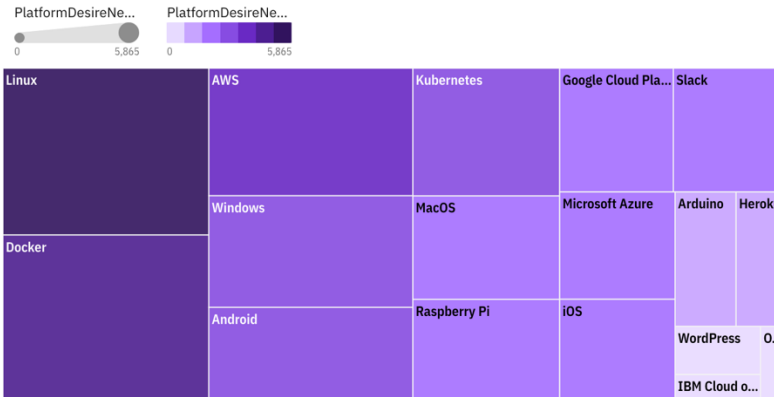
Top 10 Languages Desired Next Year



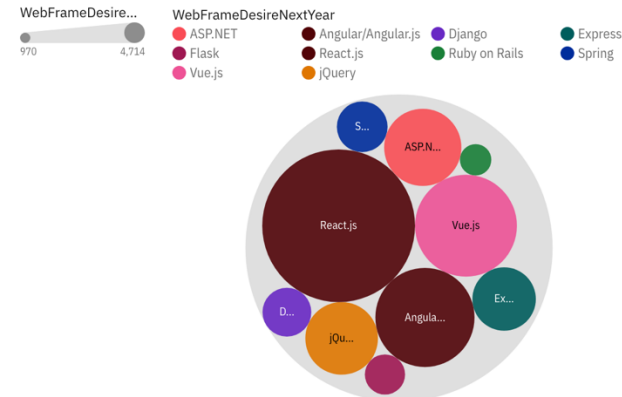
Top 10 Databases Desired Next Year



Platforms Desired Next Year



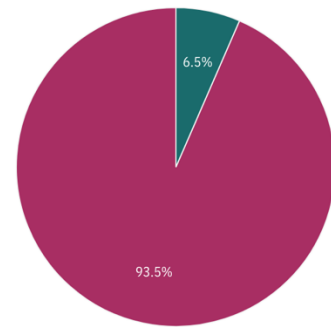
Top 10 Web Frames Desired Next Year



DASHBOARD TAB 3

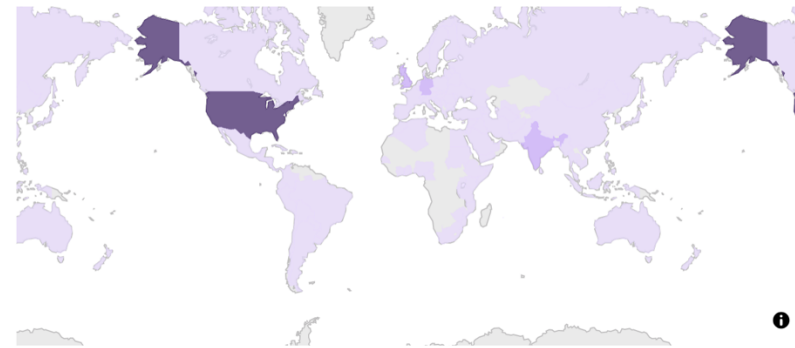
Respondent Classified by Gender

Gender
● Woman ● Man

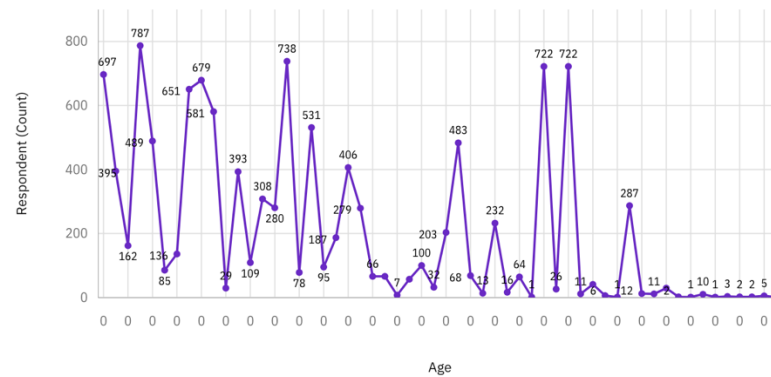


Respondent for Country regions

Respondent (Cou...
1 3,127

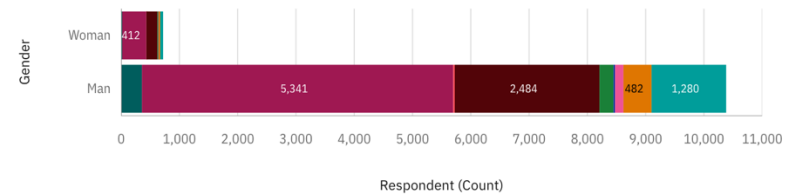


Number of Respondents by Age



Respondents by Gender with Education Levels

EdLevel
● Associate degree ● Bachelor's degree (BA, BS, B.Eng.... ● I never completed any formal edu...
● Master's degree (MA, MS, M.Eng.,... ● Other doctoral degree (Ph.D, Ed.D... ● Primary/elementary school
● Professional degree (JD, MD, etc.) ● Secondary school (e.g. American ... ● Some college/university study wit...



DISCUSSION



- As technology continues to advance, we see how developer preferences continue to change
- A majority of the respondents are male, leaving questions of how a more diverse data pool would alter preferences
- Most developers surveyed have high levels of education, which may contribute to preferences of more professional technologies vs. the self taught who would have less qualified preferences

OVERALL FINDINGS & IMPLICATIONS

Findings

- JavaScript and HTML/CSS are becoming overshadowed by Python
- The pool of databases is more diverse and leaves developers with a wide variety of competitive options
- The data surveys primarily men from the United States

Implications

- Python is growing into a staple in the programming industry for its versatility and ease of use
- Databases are fairly interchangeable and mostly based upon user preference
- The data fails to accept a broad spectrum of the totality of data programmers

CONCLUSION



- Python emerges as an exciting, new language, that developers want to learn
- Where the data succeeds in maintaining a vast age range, there is a failure to accept more female respondents and to reach a broader geographical range of respondents
- Databases and platforms are not as volatile as programming languages, therefore do not reflect as high of an importance to the respondents